1-1. 벡터 그래픽스

1-2. 래스터 그래픽스

2-1. 랜덤 스캔

2-2. 래스터 스캔

3. 10 번, $\Delta x = 4$, $\Delta y = 10$

4. (0, 1) (1, 1) (2, 2) (3, 2)

5. $\Delta x = 10$ $\Delta y = 6$ $c = \frac{7}{5}$

$$p_k = 2\Delta y x_k - 2\Delta x y_k + 2\Delta y + \Delta x(2c-1)$$

$$= 12x_k - 20y_k + 12 + 10\left(\frac{9}{5}\right)$$

$$= 12x_k - 20y_k + 30$$

$p < 0$ $p_{k+1} = p_k + 2\Delta y = p_k + 12$

$0 \le p$ $p_{k+1} = p_k + 2(\Delta y - \Delta x) = p_k - 8$

(1,2) (2,3) (3,3) (4,4)

2 -6 6 -2

6. $\Delta x=4 \quad \Delta y=10 \quad c=\frac{3}{5}$

$$y=\frac{5}{2}x-\frac{3}{2} \quad x=\frac{2}{5}y+\frac{3}{5}$$

$$p_k=2\Delta x y_k-2\Delta y x_k+2\Delta x+\Delta y(2c-1)$$

$$=8y_k-20x_k+10$$

如果 $p_{k+1}=p_k+2\Delta x=p_k+8$

否则 $p_{k+1}=p_k+2(\Delta x-\Delta y)=p_k-12$

(1,1) (2,1)

(-2) 6

7. (2,-1) $r=5$

(-2, 1)

(0,5), (2,4), -7

7-4+10

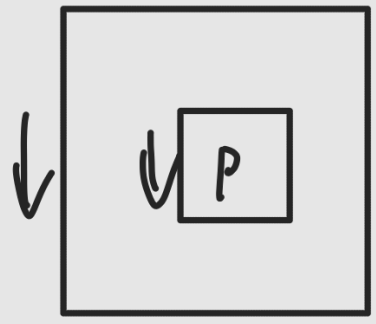
8. $p_k=3-2r$ 如果 $p_{k+1}=p_k+4x_k+6$

否则 $p_{k+1}=p_k+4(x_k-y_k)+10$

(0,5) (1,5) (2,5) (3,4)

-7 -1 9 7

9.



9-1. 안해윤다

9-2. 차H윤다

9-3. 안 해윤다

10-1. 씨드 채우기

10-2. 다각형 주사 변환 방식

11-1. EL

11-2. AEL

11-※. [N3, 6N8]?

12-1. SSAA

12-2. MSAA

12-3. FSAA

13. $(\frac{3}{4}, 0, \frac{1}{4})$

14. (4, 9)

15. (2, 3) → T(1,1) R(90) T(-1,-1)

↓

(1, 2) → (-2, 1) → (-1, 2)

↓

S(2, 1)

↓

(-2, 2)

x축 반사 ↓

(2, -2)

y방향 1 밀림

↓

(0, -2) → 이동

(2, -3)

16. $y=2x+1$ $(5,1)$

$T(-4,2)$ $T(-4,2)$

$(5,1) \to (1,3) \to (-3,5)$

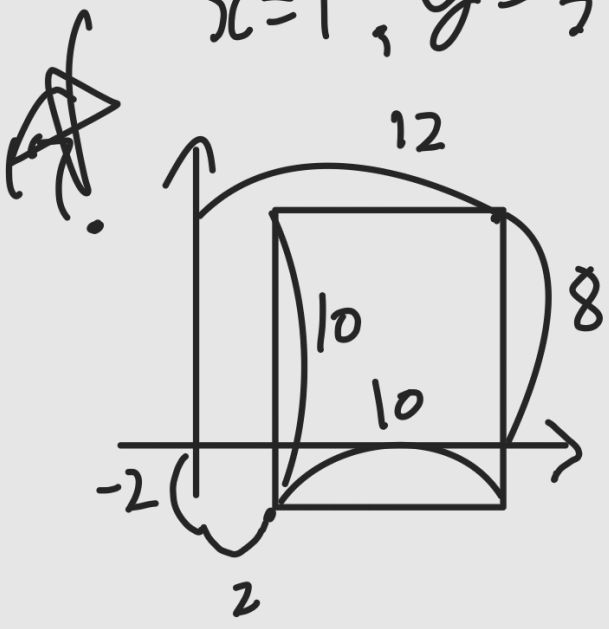
$y=\frac{1}{2}x+\frac{7}{2}$

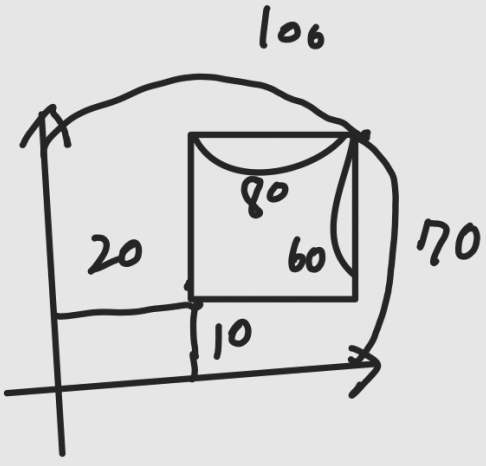
$(-3,5)$

$-x+7=4x+2$

$5x=5$

$x=1, y=3$ $C(1,3)$



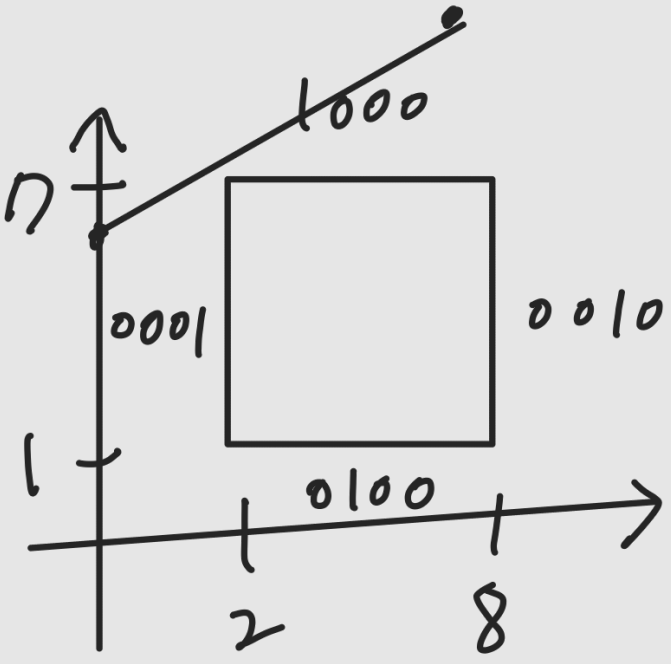


$4,3 \to T(-2,2) \to S(8,6) \to T(20,10)$

$2,5$ $16,30$ $36,40$

$(36,40)$

18.



AND는 모름다

$y=x+6$

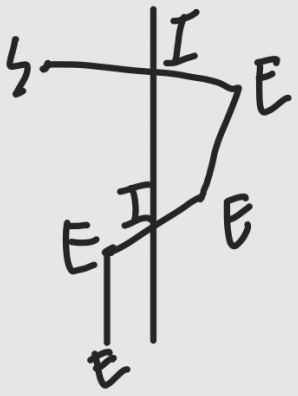
(2,8) 안지나감

전처리제거

19. $\Delta x=12$ $\Delta y=-6$ $y=-\frac{1}{2}x+6$

기억안남

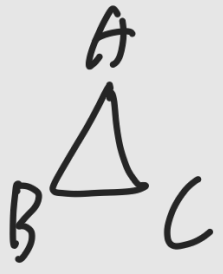
20.



I → E → E → I

21. 시계방향 기껏해야 2

22. $\overrightarrow{AB} = (0, 0, 5)$ $\overrightarrow{AC} = (8, -6, 0)$



$$\begin{vmatrix} i & j & k \\ 0 & 0 & 5 \\ 8 & -6 & 0 \end{vmatrix}$$

$(0-(-30)) - (0-40) \quad (0-0)$

$(30, 40, 0)$

$\left(\frac{3}{5}, \frac{4}{5}, 0\right)$

23. $I = 10 \times 0.2 + \frac{500}{d}\left[0.4 \times L \cdot N + 0.25 \times (R \cdot V)^2\right]$

$= 2 + 100\left[\frac{12}{50} + 0.25(R \cdot V)^2\right]$

$d = 5$

$\frac{1}{4}\left(\frac{3}{5}\right)^2 \frac{9}{25}$

$L = (1, 0, 0)$

$\frac{24}{100} + \frac{9}{100} = \frac{36}{100}$

$N = \left(\frac{3}{5}, \frac{4}{5}, 0\right)$

$R = 2(N \cdot L)N - L \quad = 38$

$= \frac{6}{5}N - L$

$= \left(\frac{18}{25}, \frac{24}{25}, 0\right) - (1, 0, 0) = \left(-\frac{7}{25}, \frac{24}{25}, 0\right)$

$V = (3, 4, 0)$

$\Downarrow$

$\left(\frac{3}{5}, \frac{4}{5}, 0\right)$

$-\frac{21}{125} + \frac{96}{125} = \frac{75}{125}$

$= \frac{3}{5}$

24-1. 빨간색

24-2. 좋아진다 그러나 힘뎃값이 면그대로

2x. 모를라